PCFileAid README

PCFileAid

PCFileAid is a Python-based utility designed to help mainframe developers, analysts, and support engineers work efficiently with copybooks, fixed-width layouts, and EBCDIC-derived data files. It provides a simple, operator-style interface for parsing, inspecting, and converting structured data used in z/OS environments.

Overview

PCFileAid is built to replicate and modernize common mainframe file inspection workflows on a Windows workstation. It supports:

- Parsing COBOL copybooks
- Generating fixed-width layout definitions
- Inspecting and validating input data
- Converting structured text into usable Python objects
- Providing a foundation for future GUI enhancements using Tkinter

The project is intended to be modular, allowing additional tools and utilities to be added over time.

Features

Copybook Parsing

- Reads COBOL copybook definitions
- Extracts field names, positions, and lengths
- Produces layout metadata for downstream processing

Layout Generation

- Converts copybook metadata into .fad or fixed-width layout formats
- Ensures alignment with mainframe-style column numbering

Data Inspection

- Reads structured text files
- Applies layout definitions to extract fields

- Supports debugging malformed or misaligned records

Tkinter Integration

- Provides a foundation for GUI-based tools
- Allows future expansion into a full PC-based FileAid utility

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File Structure

The repository currently contains:

- **pcfileaid.py** - Main program logic and Tkinter framework
- **copybook_to_layout.py** - Copybook parsing and layout generation tool
- **copybook_input_data.txt** - Sample input data for testing

Additional modules and documentation will be added as the project evolves.

Usage

Run the main program:

```
python pcfileaid.py
```

Use the layout generator:

```
python copybook_to_layout.py
```

Input files can be placed in the same directory or referenced by full path.

Requirements

- Python 3.10+
- Tkinter (included with standard Python installations)

Roadmap

Planned enhancements include:

- GUI-based copybook viewer
- EBCDIC-to-ASCII conversion utilities
- Integrated record editor
- Support for multiple layout formats
- Export to CSV, JSON, and other structured formats

License

This project is maintained privately by the author. Licensing will be defined when the project is made public.

Author

Developed by **Randy Moulton**.
For questions or contributions, please contact the repository owner.